

ex 15

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```
1 clear;
2 clc;
3 close all;
4
5 % Subcarrier Spacing values (kHz)
6 SCS_list = [15 30 60 120];
7
8 % FFT size
9 Nfft = 1024;
10
11 % Sampling frequency for 15 kHz SCS
12 fs = 15e3 * Nfft;
13
14 % Initialize symbol duration array
15 symbol_duration = zeros(1, length(SCS_list));
16
17 % Calculate OFDM symbol duration
18 for i = 1:length(SCS_list)
19
20     scs = SCS_list(i) * 1e3; % Convert kHz to Hz
21
22     % OFDM symbol duration
23     T_sym = 1 / scs;
24
25     symbol_duration(i) = T_sym;
26
27     fprintf('SCS = %d kHz -> OFDM Symbol Duration = %.5f ms\n', ...
28           SCS_list(i), T_sym * 1000);
29
30 end
31
32 % Plotting
33 figure;
34
35 plot(SCS_list, symbol_duration * 1000, '-o', ...
36      'LineWidth', 2, 'MarkerSize', 8);
37
```

Run

Save

Output

Input

SCS = 15 kHz -> OFDM Symbol Duration = 0.06667 ms
SCS = 30 kHz -> OFDM Symbol Duration = 0.03333 ms
SCS = 60 kHz -> OFDM Symbol Duration = 0.01667 ms
SCS = 120 kHz -> OFDM Symbol Duration = 0.00833 ms

[Execution complete with exit code 0]

